International Institute of Information Technology, Hyderabad

(Deemed to be University)

13.5

S24SC1.308 : The Universe Across Scales

Question cum Answer Booklet

Quiz 2

Total Marks: 15 Duration : 45 minutes

Roll No. : 2022111004 Program : CSD Date : 3 APRIL 2025

Room No. : SH2 Seat No. : _____ Invigilator Signature : _____

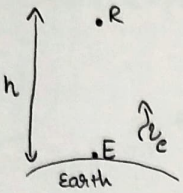
Marks Table (to be filled in by the examiner)

Question No.	Marks							Examiner
1	5							
2	3.5							
3	2							
4	3							

Instructions to the students:

1. There are four questions.
2. Place your ID card on the desk for verification by the invigilator.
3. Cheat sheets, reading materials, such as books, and electronic gadgets, like tablets, computers, mobile phones or calculators, are not allowed inside the classroom.
4. Borrowing writing materials or calculators from other students during the exam is prohibited. If any student is found indulging in malpractice or copying during the exam, the student will be given an 'F' grade for the course.

1. Prove that if if a light ray is emitted from the surface of earth and received at a height h then wavelength is redshifted. Calculate the shift in wavelength. [5]



R: Reciever E: Emitter.

Emitter emits frequency ν_e whereas R recieves at ν_R

This situation is equal to two rockets separated by distance h accelerating with acceleration g due to the Equivalence Principle.

Weak Equivalence Principle : ~~Laws of physics~~ motion of objects in free fall is same in a gravitational field and an accelerated frame.

Strong Equivalence Principle : we can construct locally inertial frame in a gravitational field such that ~~eff~~ laws of physics hold the same form as in an unaccelerated cartesian coordinate system.

Emitter emits a light ray with frequency ν_e

time taken for it to reach the reciever $= \frac{\text{distance}}{\text{speed}} = \frac{c}{h} = \frac{h}{c}$

In the mean time, R acquires a velocity due to gravity $v = gt$. $(v = u + at,\ u = 0,\ a = g)$ (downward dir)

$$v = gt = \frac{gh}{c}$$

Doppler effect states that the recieved frequency

$$\nu_R = \nu_E\left(1 - \frac{v}{c}\right) = \nu_E\left(1 - \frac{gh}{c^2}\right) = \nu_E - \nu_E\frac{gh}{c^2}$$

$$\Rightarrow \frac{\nu_R - \nu_E}{\nu_E} = \frac{-gh}{c^2}$$

The RHS is a negative quantity

$\Rightarrow \nu_R < \nu_E$

$\therefore$ The recieved freq is less than emitted freq.

$\therefore$ frequency is redshifted. — Ans.

$\parallel$

wavelength.

$$\frac{\nu_R}{\nu_E} = \frac{\lambda_E}{\lambda_R} \qquad \lambda \to \text{wavelength.}$$

$$\therefore \lambda_E = \lambda_R\left(1 - \frac{v}{c}\right)$$

$$\Rightarrow \frac{\lambda_R}{\lambda_E} = \frac{1}{1 - \frac{v}{c}} = \frac{c}{c - v}$$

$$z = \frac{\lambda_R - \lambda_E}{\lambda_E} = \frac{c}{c-v} - 1 = \frac{c - c + v}{c - v} = \frac{v}{c - v} = \frac{\frac{gh}{c}}{c - \frac{gh}{c}} = \frac{gh}{c^2 - gh}.$$

$$\therefore z = \frac{gh}{c^2 - gh} \quad \text{is the redshift value.}$$

$$1 + z = \frac{c}{c - v} = \frac{1}{\left(1 - \frac{v}{c}\right)} = \frac{\lambda_E}{\lambda_R} = \frac{\lambda_R}{\lambda_E} = \frac{\nu_E}{\nu_R}.$$

— Ans.

$$\boxed{z = \frac{\lambda_R}{\lambda_E} - 1}$$

$$\frac{\frac{1}{\lambda_R} - \frac{1}{\lambda_E}}{\frac{1}{\lambda_E}}$$

$$\frac{\lambda_E - \lambda_R}{\lambda_R}$$

$$\frac{\lambda_E}{\lambda_1}$$

$$\frac{a(h)}{a(t_e)} = 1 + z = \frac{\lambda_R}{\lambda_E}$$

2. Assume that the universe is radiation-dominated.

(a) Write down Friedmann's equations for the universe in terms of the energy density only. [0.5+0.5]

(b) Assume that the universe is spatially flat. Show that the scale factor scales as $a \propto \sqrt{t}$. Calculate the proportionality factor. [4]

Radiation dominated universe $\Rightarrow p = \rho c^2/3$
(pressure)

a) Friedman's equation $\Rightarrow \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2}$ $\rho \Rightarrow$ mass density.

acceleration equation $\Rightarrow \frac{\ddot{a}}{a} = \frac{-4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right)$

$= -\frac{4\pi G}{3}(\rho + \rho) = -\frac{8\pi G}{3}\rho$

-0.5

The energy density for radiation dominated universe $\propto \frac{1}{a^4}$ due to the loss in energy due to expansion of universe.

$\mathcal{E} = \rho c^2$ $\Rightarrow \rho = \frac{\mathcal{E}}{c^2}$

$\mathcal{E}$: Energy density; ρ: mass density.

$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\frac{\mathcal{E}}{c^2} - \frac{kc^2}{a^2}$ and $\frac{\ddot{a}}{a} = -\frac{8\pi G}{3}\frac{\mathcal{E}}{c^2}$ — Ans.

b) $k=0$ fluid equation: $\dot{\rho} + \frac{3\dot{a}}{a}(\rho + p) = 0$ $p = \rho/3$ $(c=1)$

$\dot{\rho} + \frac{\cancel{3}\dot{a}}{a}\cdot\frac{4}{\cancel{3}}\rho = 0 \Rightarrow a^4\dot{\rho} + 4a^3\dot{a}\rho = 0$

$\Rightarrow d(a^4\rho) = 0$

$\Rightarrow a^4\rho = k \Rightarrow \boxed{\rho \propto a^{-4}}$

using friedman's equation–

$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - 0$ Let $a \propto t^q$.

$\Rightarrow \dot{a} \propto t^{q-1}$ $\Rightarrow \frac{\dot{a}}{a} \propto t^{-1}$

$\Rightarrow \left(\frac{\dot{a}}{a}\right)^2 \propto t^{-2}$ [LHS]

In RHS only ρ varies with t as $\rho \propto a^{-4} \propto t^{-4q}$

since exp of t in both sides must be equal,

-3

$-2 = -4q \Rightarrow q = 1/2 \Rightarrow a \propto t^{1/2} \Rightarrow$

$\Rightarrow \boxed{a \propto \sqrt{t}}$ — Ans.

$\frac{a}{a_0} = \sqrt{\frac{t}{t_0}}$ say $a = bt^{1/2} \rightarrow \dot{a} = \frac{1}{2}bt^{-1/2} \Rightarrow \frac{\dot{a}}{a} = \frac{1}{2}t^{-1}$

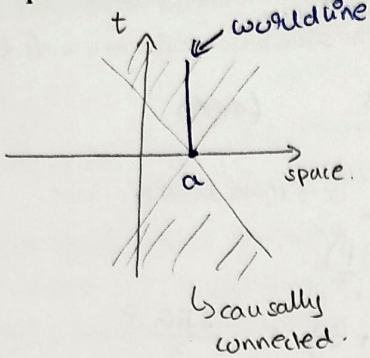
$\left(\frac{1}{2}t^{-1}\right)^2 = \frac{8\pi G}{3}\rho_0\frac{a_0^4}{a^4} = \frac{8\pi G}{3}\rho_0 a_0^4 b^{-4} t^{-2}$

$\Rightarrow b^4 = \frac{8\pi G}{3}\rho_0 a_0^4$ $\Rightarrow b = a_0\left(\frac{8\pi G}{3}\rho_0\right)^{-4}$.

$\uparrow$ proportionality factor.

— Ans.

3. Draw the worldline of an object that remains at rest at a distance a a from the origin of a light cone in a spacetime diagram. [2]



at $t=0$, object is at distance a from origin.
The particle remains at distance a even as time increases ∴ straight line parallel to y axis.
The light cone of object is also drawn and the world line lies inside the light cone with slope 0 st $-1 \le 0 \le 1$ ∴ possible.

2

4. The Friedmann-Robertson-Walker metric (or linelement)

$$ds^2 = (dt)^2 - a(t)^2\left[\frac{dr^2}{1-Kr^2} + r^2 d\theta^2\right].$$

For the Friedmann-Robertson-Walker metric in a flat universe, draw the lightcone in a spacetime diagram. [3]

Flat universe ⇒ $K=0$.

$$ds^2 = (dt)^2 - a(t)^2\left[dr^2 + r^2 d\theta^2\right]$$

r, θ → spatial coordinates.

For light like particle, $ds^2 = 0$

$$\Rightarrow (dt)^2 = a(t)^2\left(dr^2 + r^2 d\theta^2\right)$$

↑ total distance covered $= dx^2 = c \times dt^2 = dt^2$

∴ ~~$a(t)^2 = 1 \Rightarrow a(t) = \pm 1$~~.

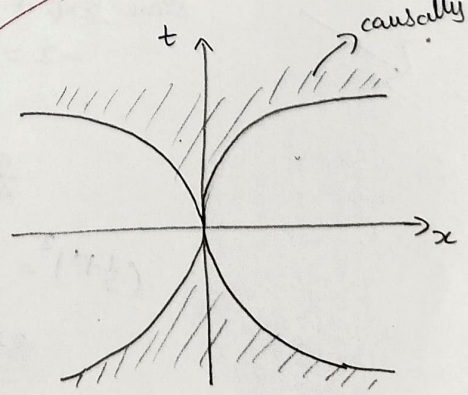
$$(dt)^2 = (a(t))^2 (dx)^2 \Rightarrow \frac{dx}{dt} = \pm\frac{1}{a(t)}$$

3

as a increases as t increases, the slope reduces and flattens

any causal connection would need slope to be b/w $-\frac{1}{a(t)}$ and $+\frac{1}{a(t)}$

as its speed $< c(1)$



light cone.